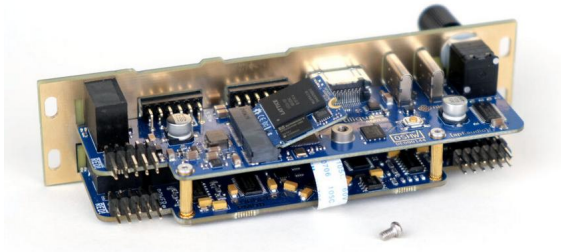


tiliqua: FPGA-based audio DSP platform

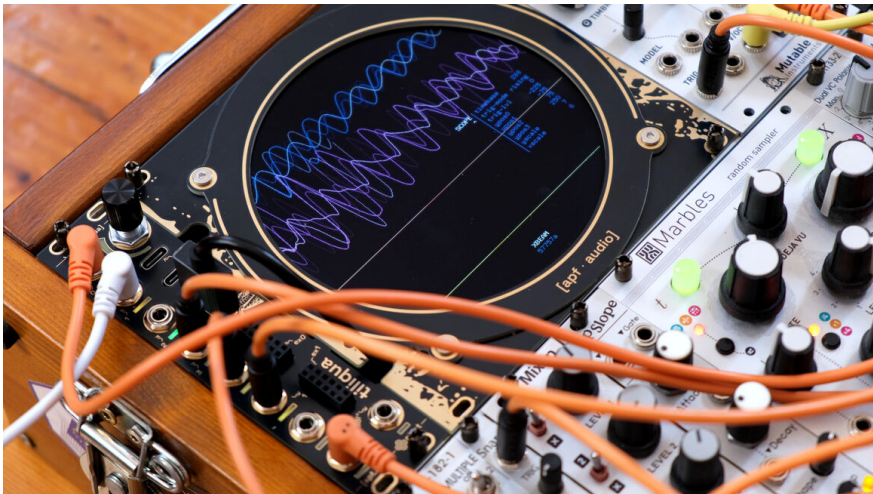
ORConf 2025

github.com/apfaudio/tiliqua

Seb Holzapfel / seb@apf.audio



Eurorack?



- ▶ Standard format for building synthesizers.
- ▶ **Eurorack ecosystem:** > 15,000 modules, > 1,000 different manufacturers.

What is tiliqua?



- ▶ Open hardware platform for audio & video synthesis (Eurorack module)
- ▶ Open gateway library of DSP components (Amaranth HDL)
- ▶ Easy to share designs – debugger built in, hot bitstream switching.
- ▶ Useful to musicians, whether they touch HDL or not.

4x IN
(192kHz, DC-coupled)

4x OUT
(192kHz, DC-coupled)



encoder

JTAG adapter

USB2 high-speed

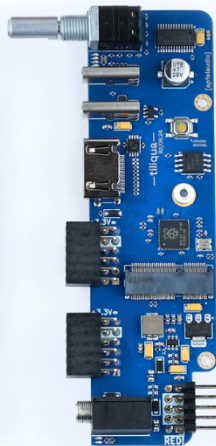
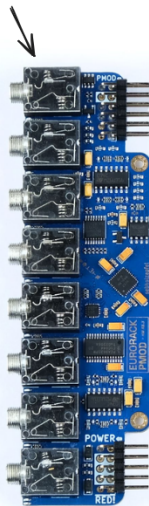
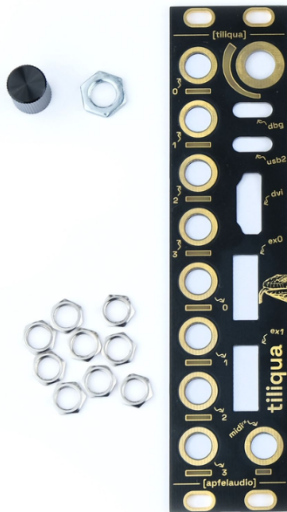
display output

2x expansion PMODs

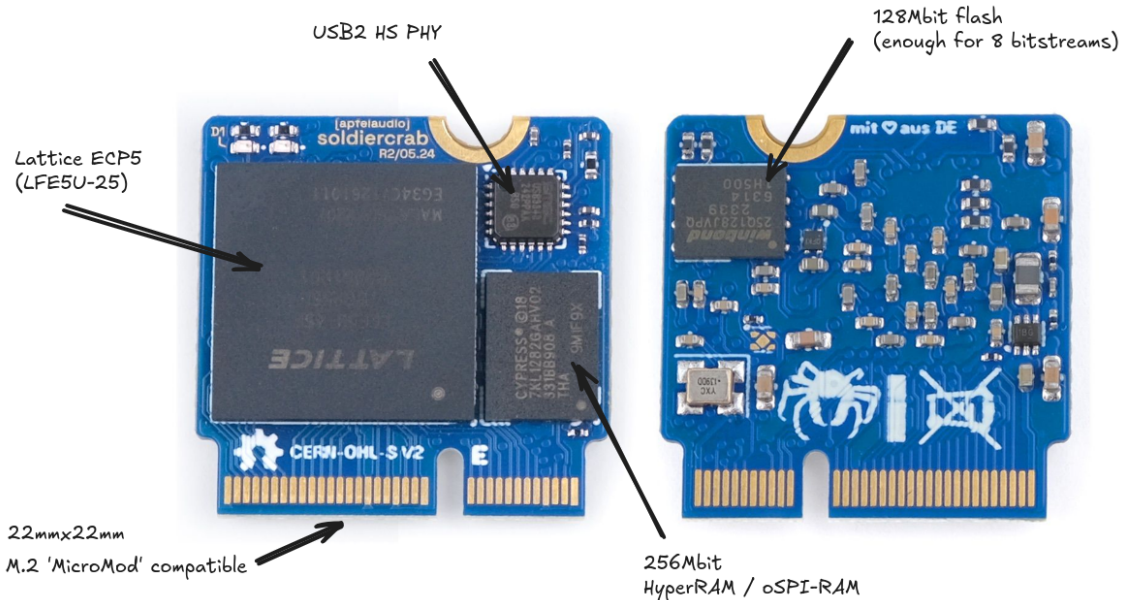
MIDI TRS IN

Audio Interface

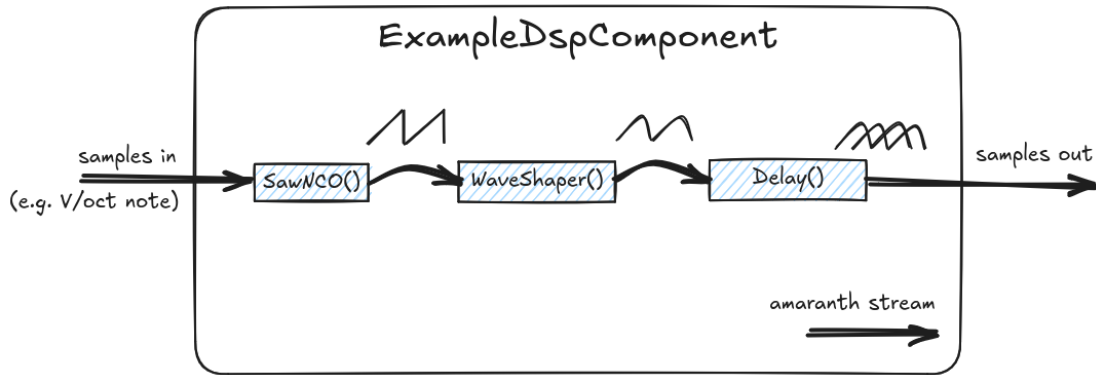
"SoldierCrab" ECP5 SoM



Tiliqua Motherboard



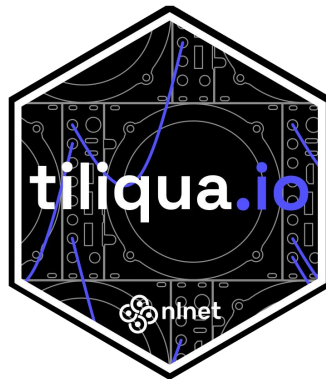
DSP pipelines, Amaranth HDL and streams



```
class ExampleDspComponent(wiring.Component):
    i: In(stream.Signature(fixed.SQ(f_width=15)))
    o: Out(stream.Signature(fixed.SQ(f_width=15)))
    def elaborate(self, platform):
        m = Module()
        # Create some DSP blocks ...
        nco      = dsp.SawNCO()
        shaper    = dsp.WaveShaper(
            lut_function=sine_osc, lut_size=128, continuous=True),
        delay     = dsp.DiffusionDelay()
        # Connect them together ...
        wiring.connect(m, flipped(self.i), nco.i)
        wiring.connect(m, nco.o, shaper.i)
        wiring.connect(m, shaper.o, delay.i)
        wiring.connect(m, delay.o, flipped(self.o))
        m.submodules += [nco, ws, rs]
        return m
```

Past year...

Past year... Funding!



Past year... Preorders!

CROWD SUPPLY

BROWSE

APPLY

ABOUT

Search

Tiliqua

A powerful, hackable FPGA-based audio multitool for Eurorack



\$46,763 raised
of \$12,000 goal

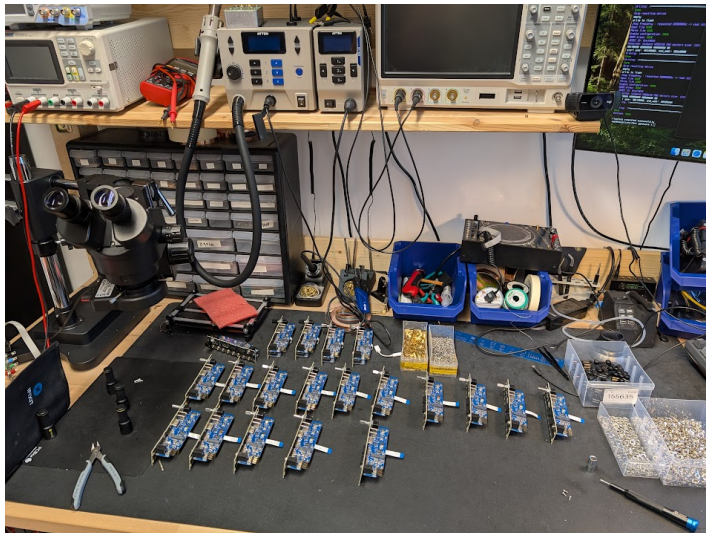
389% Funded!

6
updates

Jun 19
funded on

Available for pre-order.

Past year... Building!



Hurdles...

- ▶ **Hardware/EMC** – hairy bugs.
- ▶ **Amaranth fixed-point** – almost done, extensively tested.
- ▶ **More DSP cores** – Lots of frequency-domain stuff.
- ▶ **Dynamic pixel clocks** – hot-swap screens.
- ▶ **USB Host** – in pure gateware, based on LUNA.

Hardware / EMC: chamber



- Open source TEM Cell design (By Petteri Aimonen: essentialscrap.com/tem_cell)

Hardware / EMC: fail



Hardware / EMC: what worked

- ▶ Reroute Switchmode supplies
- ▶ Reduce FPGA pin drive strengths
- ▶ Reduce/remove split ground planes
- ▶ **Careful with long cables!**
- ▶ **Spread-spectrum PLL (on ECP5!)**



Amaranth: what is `lib.fixed`?

- ▶ **Proposed extension to Amaranth standard library**
- ▶ Fixed-point types, operations, simulation, synthesis
- ▶ Reference implementation **can be synthesized, simulated against numpy etc.!**
- ▶ **Demo:** FFT testing
- ▶ **RFC + Implementation (WIP):** github.com/amaranth-lang/amaranth/pull/1578

```
x = Signal(fixed.SQ(1, 15))
y = Signal.like(x)
m.d.sync += y.eq((x*fixed.Const(2.5)).saturate())
```

Amaranth: lib.fixed: simple filter

```
class OnePole(wiring.Component):
    i: In(stream.Signature(fixed.SQ(1, 15)))
    o: Out(stream.Signature(fixed.SQ(1, 15)))
    def __init__(self, beta=0.9):
        self.beta = beta
    def elaborate():
        m = Module()
        # Forward stream handshakes
        connect(m, flipped(self.i), self(self.o))
        # Calculate  $y = y[n-1] * \text{beta} + x * (1 - \text{beta})$ 
        a = fixed.Const(self.beta)
        b = fixed.Const(1-self.beta)
        m.d.sync += self.o.payload.eq(self.o.payload*a + self.i.payload*b)
        return m
```

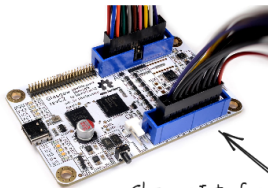
tiliqua **DSP library** - audio components

- ▶ **VCA** – Voltage controlled amplifier.
- ▶ **NCO** – Numerically controlled oscillator.
- ▶ **Trigger** – Stream hysteresis.
- ▶ **Ramp** – Synchronized ramp generator.
- ▶ **Waveshaper** – Interpolating waveshaper.
- ▶ **SVF** – Tunable state-variable filter (HPF/BPF/LPF).
- ▶ **DelayLine** – Memory-backed audio delay line.
- ▶ **PitchShift** – Granular pitch shifter.
- ▶ **MatrixMix** – Matrix mixer with tweakable coefficients.
- ▶ **FIR** – FIR filter.
- ▶ **Resample** – Rational resampler.
- ▶ **Boxcar** – Cheap averaging without multipliers.

tiliqua DSP library - (new!) frequency-domain processing

- ▶ **FFT** – forward + inverse, iterative, low area usage.
- ▶ **CORDIC** – complex/polar conversions.
- ▶ **Window** – windowing core and selectable window functions.
- ▶ **STFT** – forward + inverse, frequency-domain processing of time-domain signals.
- ▶ **SpectralEnvelope** – compute filtered power spectra over time.
- ▶ **SpectralCrossSynthesis** – compute filtered power spectra over time.

Inspiring open-hardware projects...

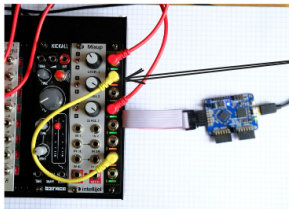


Glasgow Interface Explorer

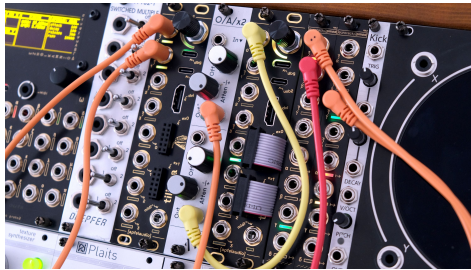


Ornament & Crime
(eurorack module)

Cynthion USB Multitool



eurorack-pmod R3
(my own project from FOSDEM23)



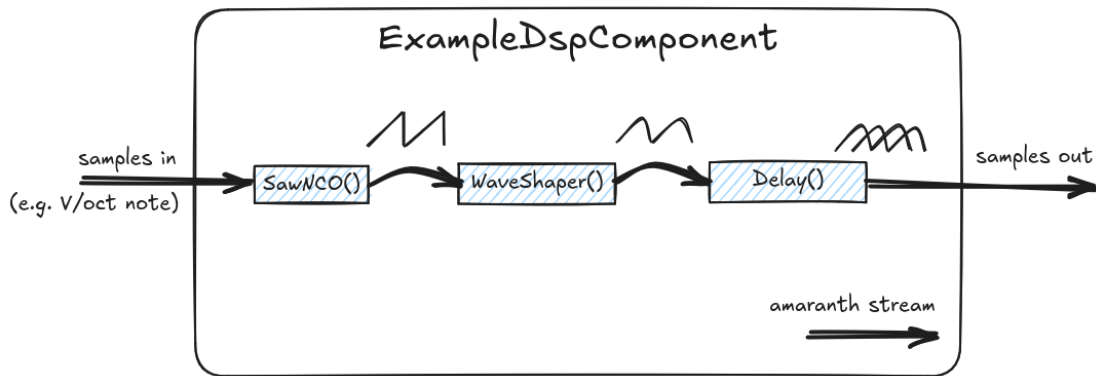
tiliqua combines –

- ▶ Easy way to get started with FPGAs & audio synthesis.
- ▶ **Eurorack**: de facto standard (audio synthesis hardware).

tiliqua goals –

- ▶ **Open hardware platform for audio & video synthesis** – Reuseable components.
- ▶ **Easy to share designs** – debugger built in, bitstream reloading mid-performance.
- ▶ **Useful to musicians, whether they touch HDL or not.**

ExampleDspComponent

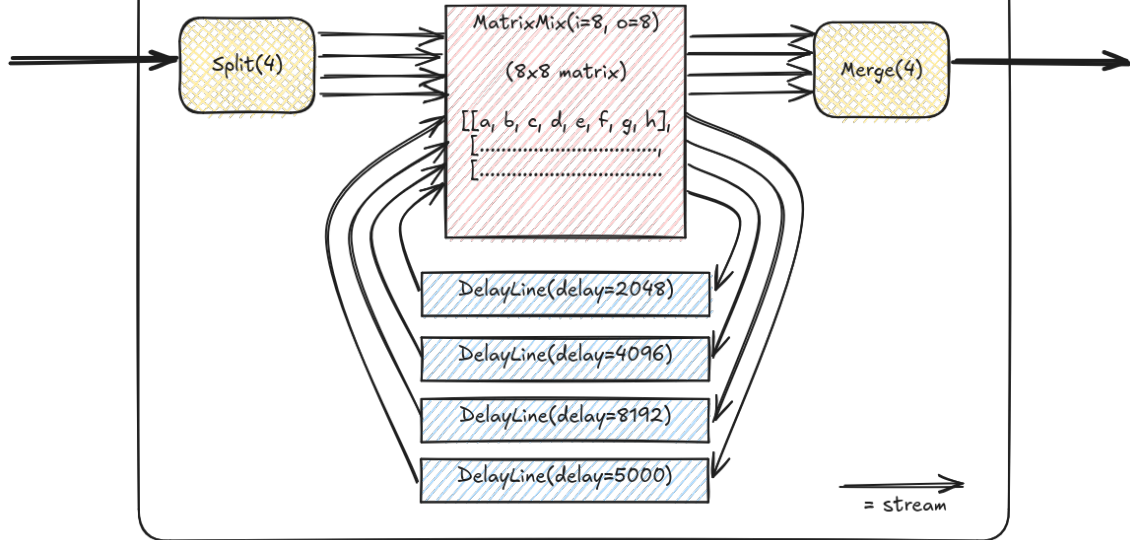


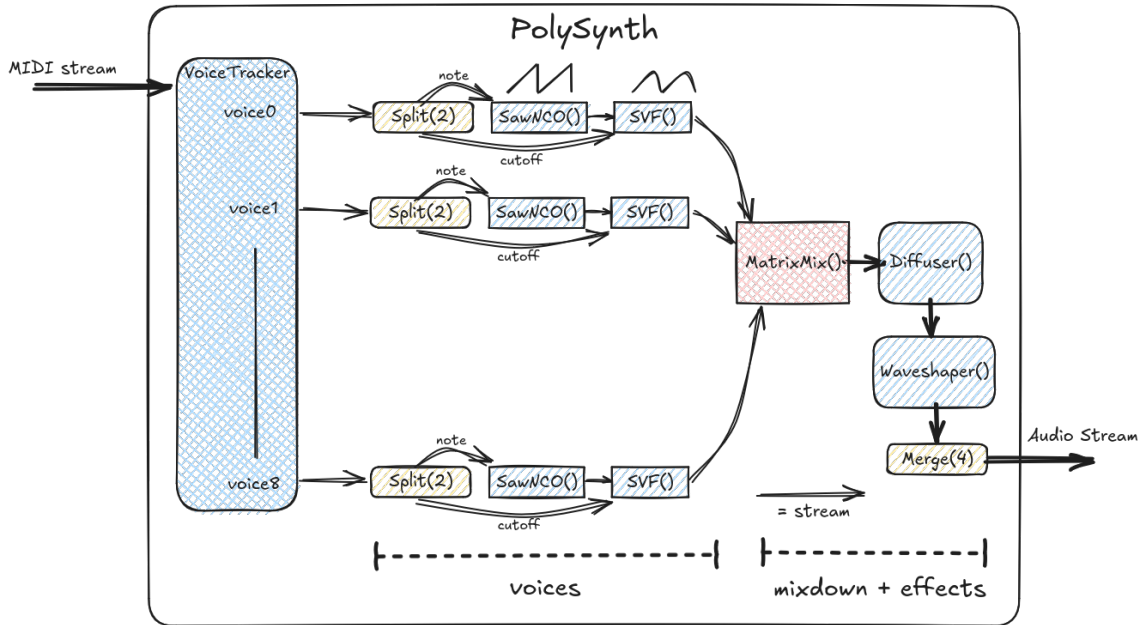
```

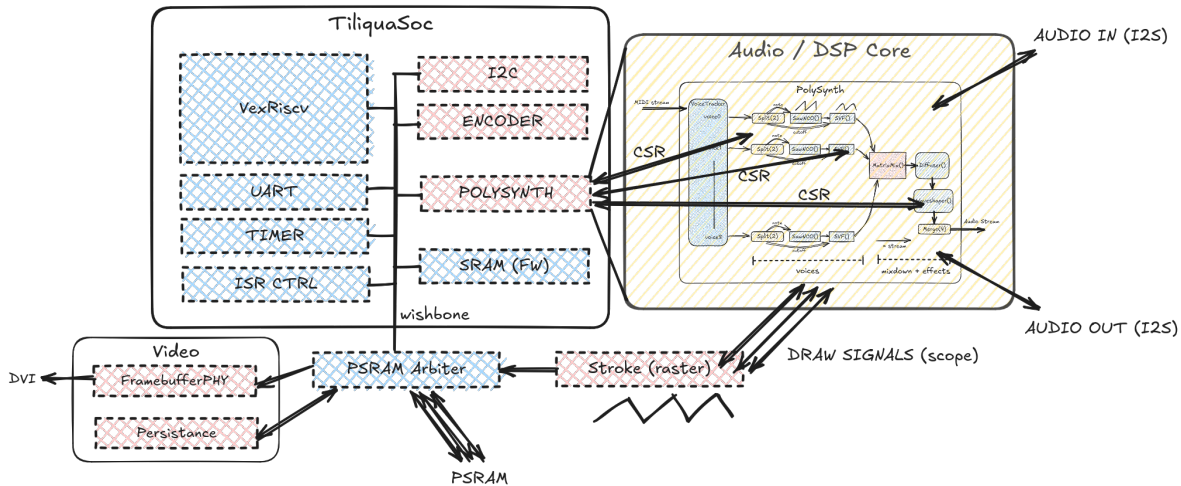
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        shaper    = dsp.WaveShaper(
            lut_function=sine_osc, lut_size=128, continuous=True),
        delay     = dsp.DiffusionDelay()
        # Connect them together ...
        wiring.connect(m, flipped(self.i), nco.i)
        wiring.connect(m, nco.o, shaper.i)
        wiring.connect(m, shaper.o, delay.i)
        wiring.connect(m, delay.o, flipped(self.o))
        m.submodules += [nco, ws, rs]
        return m

```

Diffusion Delay





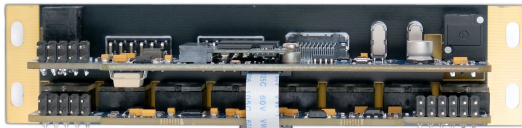


Demo - example bitstreams



- ▶ **xbeam** – 4-ch vectorscope, oscilloscope.
- ▶ **polysyn** – 8-voice polyphonic synth (touch or MIDI control).
- ▶ **sidula** – C64 SID emulation to modular.
- ▶ **usb-sound** – 4 IN, 4 OUT USB sound card.
- ▶ **quadnco** – 4ch oversampling wavetable oscillator.
- ▶ **diffuser** – Diffusion delay effect.

Thank you for listening!



- ▶ contact: `seb@apf.audio`
- ▶ repository: `github.com/apfaudio/tiliqua`
- ▶ mailing list: `apf.audio`

(almost) shipping!

- ▶ `crowdsupply.com/apfaudio/tiliqua`